# Varnikumar Rajendrakumar Patel

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## EXPERIENCE

**FOSSEE, IIT Bombay** May, 2026 - Jul, 2026
**Software Development Fellow - Osdag Web** Mumbai, India (Remote)

- Integrated LLM workflows into legacy Osdag codebase to enable context-aware engineering assistance.
- Refactored core React interface components and dynamic SVG algorithms to optimize app responsiveness.
- Implemented 10+ feature enhancements, improving application functionality and overall user workflows.
- Collaborated with 5+ engineers to thoroughly validate and deliver production-ready application updates.

**Mittal Alliance Industries Private Limited** May, 2025 - Jan, 2026
**Full Stack Web Developer Intern** Indore, India (Hybrid)

- Delivered interactive interfaces for 10+ websites using common UX patterns for a pixel-perfect UI.
- Contributed to 3 major projects with the backend team utilizing Node, Express, MongoDB, and Firebase.
- Coordinated with clients and 4 programmers to align base project requirements and ensure timely delivery.
- Led a team of 6 developers, closely reviewing 50+ logic updates to maintain exceptional code quality.

## PROJECTS

**CompileVerse, Full-Stack Cloud IDE & AI Code Companion** Jun, 2026 – Sep, 2026
**Tech Stack:** React.js, Node.js, Express.js, LangChain, Groq API, RAG, Docker Live Link

- Engineered a scalable containerized IDE supporting 4 major languages using React, Node.js, and Docker.
- Architected a secure Node.js sandbox enforcing 5s timeouts and 1MB memory limits to block crashes.
- Designed a batch test runner with Big-O complexity analysis, reducing execution latency to 360ms.
- Integrated a LangChain RAG pipeline with Groq LLaMA-3.3 and Voice AI for real-time code tutoring.

**ArchMind - AI-Powered System Design Learning Platform** Jul, 2026 - Aug, 2026
**Tech Stack:** Next.js, MongoDB, Groq LLM API Live Link

- Built a full-stack AI System Design platform with HLD, LLD, and AI Mentor workflows using Next.js.
- Developed reusable UX components via a custom React plugin, parsing 50+ visual elements into JSON.
- Integrated Groq LLM APIs for automated architecture reviews and Terraform IaC generation in under 3s.
- Built a secure dashboard using NextAuth/JWT, achieving 95+ Lighthouse performance and <1s load time.

## TECHNICAL SKILLS

- **Programming Languages:** Python, Java (DSA, OOPS), JavaScript, TypeScript,
- **Frontend Technologies:** React.js, Next.js, Tailwind CSS, HTML, CSS, Bootstrap, Framer Motion
- **Backend & Databases:** Node.js, Express.js, RESTful APIs, Redis, Kafka, MongoDB, Firebase
- **Generative AI & Tools:** LLMs, Prompt Engineering, RAG, LangChain, Docker, Git, CI/CD, Postman

## ACHIEVEMENTS & LEADERSHIP

- Twice selected for **Amazon ML Summer School (2025 & 2026)**, ranking among top 3,000 candidates.
- Earned LeetCode Knight badge with a peak **Contest Rating of 2032**, ranking in the **top 2.22% globally**.
- Served as Technical Lead of a 5-member team at Ramanujan Club, driving key initiatives and mentoring.

## EDUCATION

**Vellore Institute of Technology** Jun, 2023 - May, 2027
B.Tech in Computer Science & Engineering (Artificial Intelligence and Machine Learning) **CGPA**: 8.06 / 10.00